Can Light Tell Whether Spacetime Is Complex?

Nine directions and one undetermined scalar in Complex Relativity

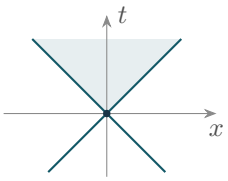
Gauss-K[1]

6 November 2026

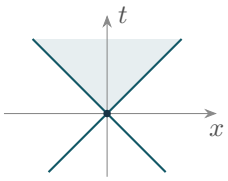
Abstract. Suppose a complex metric $\gamma = g + \mathrm{i}b$ shares every real lightlike direction of a Lorentzian metric g. What freedom does that leave for b? In four dimensions, evaluating b along six axial directions and three diagonals suffices to establish $b = \chi g$. The value of χ drops out of every pointwise cone test, and under the momentum postulate of Complex Relativity it also drops out of the free mass shell. Gradients of χ have a different status. For one choice of gravitational action they carry stress and contribute to the focusing of light. We calculate the angular-diameter distance in a universe filled by this scalar and compare it with a matter interaction whose strength depends on the response parameter C.

CR manuscripts: github.com/Gauss-K/Complex-Relativity

Suppose two colleagues hand you candidate spacetime metrics. The first writes g, the second $(1 + \mathrm{i}/2)g$. Both use the same real clocks. You may send a flash in any direction and record which tangents at a single event have zero interval. How would you tell which colleague wrote which metric?



$\gamma_A = g$



$\gamma_B = (1 + \mathrm{i}/2)g$

The rays have identical slopes. For any real tangent v with $g(v, v) = 0$, the second squared interval is $(1 + \mathrm{i}/2)\, 0 = 0$. We could replace $1/2$ by any real number and draw exactly the same lines.

Now suppose the imaginary part b is given as a symmetric matrix with ten independent entries. Could those entries cancel along every ray even if b were more complicated than a multiple of g? The coordinate axes are an obvious place to start. A single off-diagonal entry can pass all six tests.

[1]Contact: complex.relativity@outlook.com

1 Clocks, light rays and the real slice

An observer sends a light pulse to a nearby mirror at proper time τ_- and receives the reflection at τ_+. In local radar coordinates the reflection occurs at time $(\tau_+ + \tau_-)/2$ and distance $(\tau_+ - \tau_-)/2$, in units with the speed of light equal to one. Mirrors in different directions give different paths for the same clock to time. At a point, knowing the entire light cone fixes a real Lorentzian metric up to scale. Assigning a proper time to a specified timelike displacement fixes that remaining scale. The distinction between the geometry supplied by light and the information supplied by clocks figures in the work of Ehlers, Pirani and Schild [2].

Complex Relativity (CR) [1] places this observer in a spacetime $\mathcal{X}$ of four complex dimensions, equipped with a symmetric complex bilinear metric. Physical events lie on a four-dimensional real slice M. The slice is *maximally real*, meaning that at a point x every complex tangent has a unique expression $v + \mathrm{i}w$ with $v, w \in T_xM$. On real displacements the metric restricts to

$$\gamma = g + \mathrm{i}b, \qquad \gamma(v, v) = g(v, v) + \mathrm{i}b(v, v). \tag{1}$$

Here g is the metric of the observer's real clocks, with signature $(- + ++)$, and b is a second real symmetric form. The connection between this complex geometry and the physics of clocks, light rays and free particles rests on two foundational postulates:

P1 Freely falling observers have equivalent local inertial descriptions and use real proper time in a time-oriented Lorentzian geometry. Every real lightlike tangent of g also has zero squared interval for γ. Equivalently, $g(v, v) = 0$ implies $b(v, v) = 0$.

P2 The momentum of a free particle is a real covector, its invariant mass is real and nonnegative, and its motion is canonical on the real mass shell. Momentum extends complex-linearly from the slice. The complex quadratic mass shell must describe the same free states and recover the ordinary shell in the flat real reference geometry.

The last clause of P1 about the complex interval is an assumption in its own right. Take flat g and $b = \beta\,\mathrm{d}t^2$, with $\beta \neq 0$. Clocks and signals coupled only to g would give their usual readings, yet the real lightlike tangent $(1, \mathbf{n})$, $|\mathbf{n}| = 1$, has complex squared interval $\mathrm{i}\beta$. For a proposed metric we can check P1 by substitution. Turning that calculation into an experiment would require a way of reading out b.

2 Turn the ray through forty-five degrees

In an orthonormal frame at the event, write a future lightlike tangent as $v(\mathbf{n}) = (1, \mathbf{n})$ with $|\mathbf{n}| = 1$. The choice $v^0 = 1$ fixes its normalization. The imaginary part of its squared interval is

$$d(\mathbf{n}) = b(v(\mathbf{n}), v(\mathbf{n})) = b_{00} + 2b_{0i}n_i + b_{ij}n_i n_j. \tag{2}$$

P1 says that d vanishes everywhere on the unit sphere.

Take a matrix whose only nonzero entries are $b_{12} = b_{21} = \beta$. A ray along any axis, in either direction, gives $d = 0$, however large β is. For a ray turned through an angle θ in the $1, 2$ plane, Eq. (2) gives

$$\mathbf{n} = (\cos\theta, \sin\theta, 0), \qquad d(\mathbf{n}) = 2\beta\cos\theta\sin\theta = \beta\sin(2\theta). \tag{3}$$

On the diagonal at 45° the residual is β. The two coordinates enter with equal weight there. On either axis one of them vanishes, taking the cross term with it.

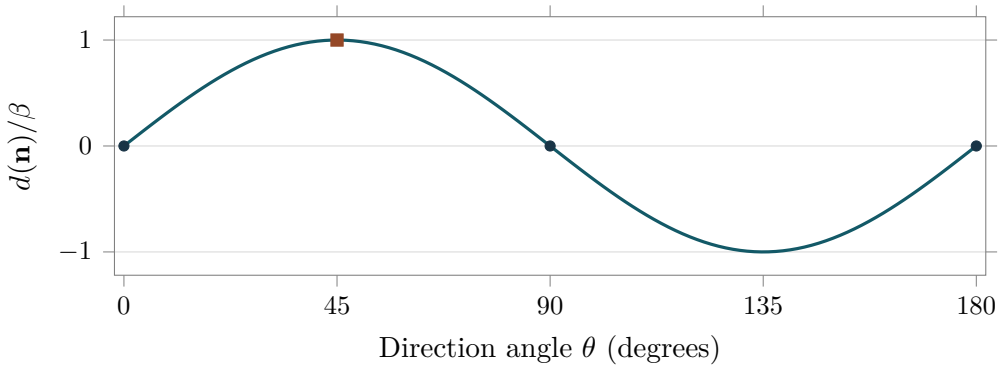


Figure 1. The circles mark rays along the axes in the $1, 2$ plane. The square marks the diagonal where the residual equals the off-diagonal matrix entry b_{12}. The vertical scale is in units of $\beta \neq 0$.

The quadratic dependence on the tangent is what makes a finite test possible. Let $B = (b_{ij})$ be the spatial block. On the sphere, d has a constant part $b_{00} + \operatorname{tr} B/3$, a dipole $2b_{0i}n_i$, and a quadrupole $(b_{ij} - \delta_{ij}\operatorname{tr} B/3)n_i n_j$. These carry one, three and five independent coefficients. The cross term in Figure 1 occupies one of the five quadrupole components. Higher angular harmonics would require a more general response than a quadratic interval.

3 Nine directions and the remaining scalar

Take the six axial directions and add one diagonal in each coordinate plane. To see what these nine evaluations determine, denote the three unit axes by e_i and suppose that all nine values of d vanish.

Directions	Evaluation	Consequence
$\pm e_i$	$d(e_i) - d(-e_i)$	$b_{0i} = 0$
$\pm e_i$	$d(e_i) + d(-e_i)$	$b_{ii} = -b_{00}$
$(e_i + e_j)/\sqrt{2},\ i < j$	$d((e_i + e_j)/\sqrt{2})$ after the axial tests	$b_{ij} = 0$

On an axis $d(\pm e_i) = b_{00} \pm 2b_{0i} + b_{ii}$, which gives the first two rows. After those conditions are imposed, only b_{ij} survives on the i, j diagonal. This leaves a single independent component. In tensor form the result is

$$b = \chi g, \qquad \chi = \tfrac{1}{4} g^{\mu\nu} b_{\mu\nu}, \qquad \boxed{\gamma = (1 + \mathrm{i}\chi) g}\,. \tag{4}$$

Conversely $b = \chi g$ vanishes on every real lightlike tangent. Since $1 + \mathrm{i}\chi \neq 0$, g and γ have exactly the same real light cone.

Nine is minimal for the stated test. A symmetric 4×4 matrix has ten independent entries. Each direction gives one homogeneous linear condition on them. Eight directions leave a kernel of dimension at least two. The nine in the table leave precisely the line spanned by g. This is a pointwise statement. On a smooth real slice the trace in Eq. (4) makes χ a smooth field whose value can vary from event to event.

Finite accuracy. The table also tells us how errors in the residuals enter the inferred components. Let $d_i^{\pm} = d(\pm e_i)$ and $d_{ij} = d((e_i + e_j)/\sqrt{2})$. Write $a_i = (d_i^+ + d_i^-)/2$ and $w_i = (d_i^+ - d_i^-)/4$. Direct substitution in Eq. (2) gives

$$b_{00} + b_{ii} = a_i, \quad b_{0i} = w_i, \quad b_{ij} = d_{ij} - \tfrac{1}{2}(a_i + a_j) - \sqrt{2}\,(w_i + w_j) \quad (i < j). \tag{5}$$

If all nine residuals have magnitude at most ε, then $|a_i| \leq \varepsilon$ and $|w_i| \leq \varepsilon/2$. For b_{ij}, the coefficients of d_i^+ and d_j^+ are each $-(1+\sqrt{2})/4$. Those of d_i^- and d_j^- are each $(\sqrt{2}-1)/4$, and that of d_{ij} is one. Their absolute values sum to $1 + \sqrt{2}$, giving

$$|b_{ij}| \leq (1 + \sqrt{2})\varepsilon, \qquad i < j. \tag{6}$$

The bound is attained by assigning each of these five residuals magnitude ε and the sign of its coefficient. Rank nine guarantees a symmetric b with those data. The numerical bound refers to the chosen frame and to $v^0 = 1$. Even with $\varepsilon = 0$, the value of χ is free.

4 What a massive particle adds

Light tests the metric on tangents of zero squared length. A massive particle comes with a scale of its own, its rest mass m. Its momentum might therefore seem a promising way to determine χ. In CR the answer depends on how the measured momentum enters the complex mass shell.

Let u be the observer's future unit four-velocity. Under P2 the momentum p is a real covector, and the observer measures the energy $E = -p(u)$. On the positive-energy branch the free constraint reads

$$\mathcal{Q}_{\mathrm{R}} = g^{\mu\nu} p_\mu p_\nu + m^2 = 0. \tag{7}$$

Maximal reality of the slice gives p a unique complex-linear extension, $p^{\mathbb{C}}(v + \mathrm{i}w) = p(v) + \mathrm{i}p(w)$. Its pairing with a real displacement is the familiar $p_\mu \mathrm{d}x^\mu$. Write $F = 1 + \mathrm{i}\chi$. Since $\gamma^{-1} = F^{-1} g^{-1}$, contraction with the complex inverse metric multiplies the kinetic term $g^{\mu\nu} p_\mu p_\nu$ by F^{-1}.

Suppose the accompanying mass term is $A(x)m^2$, with A independent of momentum. On the real shell the proposed complex constraint equals $(A - F^{-1})m^2$. For $m > 0$ it vanishes only if $A = F^{-1}$. The complex shell can then be written

$$\mathcal{Q}_{\mathbb{C}} = F^{-1} \mathcal{Q}_{\mathrm{R}}, \qquad \mathrm{Re}\, \mathcal{Q}_{\mathbb{C}} = \frac{\mathcal{Q}_{\mathrm{R}}}{1 + \chi^2}, \qquad \mathrm{Im}\, \mathcal{Q}_{\mathbb{C}} = -\chi\, \mathrm{Re}\, \mathcal{Q}_{\mathbb{C}}. \tag{8}$$

The two parts supply a single independent real condition on the regular shell. For $m = 0$ the mass coefficient drops out. The massless shell is regular away from $p = 0$. Setting $A = 1$ in the complex shell would impose $\chi m^2 = 0$, excluding massive particles wherever $\chi \neq 0$.

The converse shows how much information a massive shell carries. Suppose a quadratic form h on covectors takes the value $-Am^2$ everywhere on the future shell of one positive mass, with the same A in every direction. Then $h - Ag^{-1}$ vanishes on that shell. By rescaling p, it vanishes throughout the open future timelike cone. A quadratic polynomial zero on an open set is identically zero, hence $h = Ag^{-1}$.

For free motion the real canonical action is $\int (p_\mu \dot{x}^\mu - e\mathcal{Q}_{\mathrm{R}}/2)\mathrm{d}\tau$, with e a real multiplier. Its trajectories are geodesics of g. At fixed g and m, changing χ leaves both the trajectory and the local dispersion relation $E^2 = |\mathbf{p}|^2 + m^2$ intact. The free particle fixes the relative normalization of the kinetic and mass terms. The value of χ itself is still undetermined.

5 When the scalar gravitates

So far we could assign χ a different value at every event and satisfy the same pointwise condition. To calculate the energy of such a profile, we have to choose a gravitational action. Consider the real part of the complex Einstein–Hilbert action on the allowed metrics,

$$S_g = \frac{M_{\mathrm{P}}^2}{2}\,\mathrm{Re}\int_M \mathrm{d}^4x\,\sqrt{-\det\gamma}\,R[\gamma], \qquad \gamma = Fg, \quad F = 1 + \mathrm{i}\chi. \tag{9}$$

with M_{P} the reduced Planck mass. We insert $\gamma = (1 + \mathrm{i}\chi)g$ before varying the action. The independent real fields are g and χ, so the allowed variations have the form $\delta\gamma = F\delta g + \mathrm{i}g\delta\chi$. Choosing Eq. (9) is the extra assumption about dynamics in this calculation.

In four dimensions, the branch of the volume element continuous from $F = 1$ satisfies $\sqrt{-\det\gamma} = F^2\sqrt{-g}$. The conformal transformation of the scalar curvature gives

$$\sqrt{-\det\gamma}\,R[\gamma] = \sqrt{-g}\left(FR[g] - 3\Box_g F + \frac{3}{2F}(\nabla F)^2\right). \tag{10}$$

The contractions on the right use g. The $\Box_g F$ term is a divergence. With the boundary terms treated accordingly, the remaining gradient term has real part $\mathrm{Re}[(\nabla F)^2/F] = -(\nabla\chi)^2/(1+\chi^2)$. A change of variable puts its kinetic term in canonical form,

$$f = \sqrt{\tfrac{3}{2}}\,M_{\mathrm{P}}, \qquad \phi = f\ \mathrm{arsinh}\,\chi, \tag{11}$$

$$S_g = \int_M \mathrm{d}^4x\,\sqrt{-g}\left[\frac{M_{\mathrm{P}}^2}{2}R[g] - \frac{1}{2}(\nabla\phi)^2\right]. \tag{12}$$

In these variables the theory is Einstein gravity coupled to a real massless scalar. A constant ϕ contributes zero stress. A profile with a gradient can curve the metric read by clocks.

In the absence of other matter, the trace-reversed Einstein equation is $M_{\mathrm{P}}^2 R_{\mu\nu} = \partial_\mu\phi\,\partial_\nu\phi$. Contract it with the tangent $k^\mu = \mathrm{d}x^\mu/\mathrm{d}\lambda$ of a light ray to obtain

$$R_{\mu\nu}k^\mu k^\nu = \frac{(k^\mu\partial_\mu\phi)^2}{M_{\mathrm{P}}^2} = \frac{3}{2}\frac{(k^\mu\partial_\mu\chi)^2}{1+\chi^2}. \tag{13}$$

The numerator is the square of $\mathrm{d}\chi/\mathrm{d}\lambda$. A ray running along a surface of constant χ receives zero Ricci focusing from this scalar. A ray along which χ changes receives a positive contribution. The geometry offers a way to see the difference by following neighboring rays and measuring their separation.

6 The width of a light beam

Consider a narrow, twist-free beam of light. Let $\mathcal{A}$ be its transverse area, with the rays affinely parametrized by λ. Its fractional rate of expansion is $\vartheta = \mathrm{d}\log\mathcal{A}/\mathrm{d}\lambda$. The null Raychaudhuri equation is [3]

$$\frac{\mathrm{d}\vartheta}{\mathrm{d}\lambda} = -\frac{\vartheta^2}{2} - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu, \tag{14}$$

where $\sigma_{\mu\nu}$ is the shear on the transverse screen. Positive $R_{\mu\nu}k^\mu k^\nu$ makes ϑ decrease faster. The beam can still grow in area while this happens.

There is an exact example in a spatially flat universe whose only matter is the homogeneous scalar in Eq. (12). One expanding solution at $t > 0$ is

$$\begin{aligned} \mathrm{d}s_g^2 &= -\mathrm{d}t^2 + a^2(t)\,\mathrm{d}\boldsymbol{x}^2, \qquad a(t) = a_*(t/t_*)^{1/3}, \\ \phi(t) &= \phi_* + \sqrt{2/3}\,M_{\mathrm{P}}\log(t/t_*). \end{aligned} \tag{15}$$

Here $a_*, t_* > 0$ and ϕ_* set the reference values. With $H = \dot{a}/a = 1/(3t)$ and $\dot{\phi} = \sqrt{2/3}\,M_{\mathrm{P}}/t$, both $3M_{\mathrm{P}}^2H^2 = \dot{\phi}^2/2$ and $\ddot{\phi} + 3H\dot{\phi} = 0$ hold. A comoving observer measures light frequency $\omega = k^t$. Substitution into Eq. (13) gives

$$R_{\mu\nu}k^\mu k^\nu = \frac{2\omega^2}{3t^2}. \tag{16}$$

One can check the coefficient directly from the metric. For a radial ray $k^r = \omega/a$, and the required components are $R_{tt} = -3\ddot{a}/a$ and $R_{rr} = a\ddot{a} + 2\dot{a}^2$. Their contraction with $k^\mu k^\nu$ reproduces Eq. (16).

A source of known transverse size seen at a small angle gives the angular-diameter distance D_A, its size divided by that angle. For light emitted at t_e and observed at t_o, the radial light path gives $D_A = a(t_e)\int_{t_e}^{t_o}\mathrm{d}t/a(t)$. In terms of the redshift $1 + z = a(t_o)/a(t_e)$ and $H_o = 1/(3t_o)$, this becomes

$$H_oD_A(z) = \tfrac{1}{2}\left[(1+z)^{-1} - (1+z)^{-3}\right]. \tag{17}$$

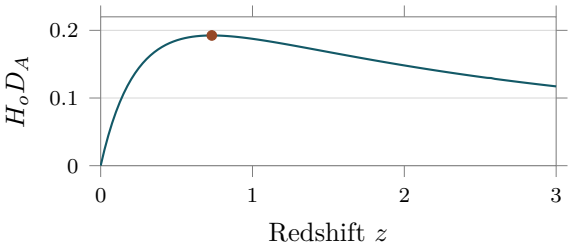


Figure 2. In this scalar-filled universe D_A initially grows as z/H_o and reaches a maximum at $z = \sqrt{3} - 1$. The distance samples the geometry along the light path. Einstein gravity with an ordinary massless scalar gives exactly the same curve.

7 Coupling the scalar to matter

The beam in Figure 2 can be described entirely in the language of a real scalar field. A matter coupling gives us a second calculation to compare with gravity. Write two real fields as $Z = q_1 + \mathrm{i}q_2$. One response law considered in CR [1] assigns the following quadratic form to their field differentials,

$$\mathrm{d}\ell_C^2 = \mathrm{Re}\Big[F(\mathrm{d}Z)^2\Big] + C|F|\,|\mathrm{d}Z|^2, \qquad |F| = \sqrt{1+\chi^2}. \tag{18}$$

The dimensionless parameter C specifies the strength of the second term. This is a metric on the space of field values. Spacetime derivatives in the kinetic action are contracted with g. In real components the quadratic form is

$$(C|F|+1)\mathrm{d}q_1^2 - 2\chi\,\mathrm{d}q_1\,\mathrm{d}q_2 + (C|F|-1)\mathrm{d}q_2^2. \tag{19}$$

Its eigenvalues are $(C+1)|F|$ and $(C-1)|F|$. Both kinetic directions are positive for $C > 1$. At $C = 1$ one eigenvalue vanishes.

For a scattering experiment within the model, expand about $\chi = \phi = 0$, set the potential to zero, and work on fixed Minkowski spacetime. The canonically normalized matter fields are $U = \sqrt{C+1}\,q_1$ and $V = \sqrt{C-1}\,q_2$. With the usual kinetic prefactor $-1/2$, the cross term in Eq. (19) gives the cubic interaction

$$\mathcal{L}_3 = \lambda_C \phi\,\partial_\mu U\,\partial^\mu V, \qquad \lambda_C = \frac{1}{f\sqrt{C^2-1}}. \tag{20}$$

In $UU \to VV$, exchange of ϕ in the t and u channels gives $-\lambda_C^2(t+u)/4$. For massless external particles $s+t+u = 0$, with $s > 0$ in the physical channel. The leading tree amplitude is therefore

$$\mathcal{M}^{(2)}(UU \to VV) = \frac{s}{4f^2(C^2-1)}. \tag{21}$$

The coefficient of s is also the sectional curvature of the U, V plane at the vacuum in the target metric $\mathrm{d}\phi^2 + \mathrm{d}\ell_C^2$. This gives a check using the field-space description of scalar scattering [4].

At the same f, $C = 2$ gives a slope of $1/(12f^2)$, while $C = 3$ gives $1/(32f^2)$. The ratio is $8/3$. Both models give the same spacetime light cone. The difference in slopes comes from the particular matter law in Eq. (18), for which C is a continuous parameter.

8 Comparing gravity with matter

Return to the two sketches on the first page. Knowing all the real directions in which light travels fixes nine combinations of the imaginary metric at one event. A constant imaginary multiple of g escapes every one of those tests. If the multiple varies along a ray, the restricted action in Eq. (9) assigns it stress, and the evolution of the beam becomes sensitive to its gradient. An ordinary massless scalar gives the same gravitational signal.

The scale $f = \sqrt{3/2}\, M_{\rm P}$ that normalizes the gravitational scalar in Eq. (11) also sets the size of the amplitude in Eq. (21). If C could be determined independently, scattering would give a second value for f to compare with gravity. That comparison requires U, V states that can be prepared and detected, as well as a calculation that includes graviton exchange. At this normalization it contributes at the same derivative order as scalar exchange. Agreement would test a specific relation between the gravitational and matter couplings. A real scalar theory with the same couplings would satisfy it too.

9 Further reading

The project manuscripts develop different parts of this argument. They are collected at github.com/Gauss-K/Complex-Relativity.

A Curious Reader's Guide (English) and A Curious Reader's Guide (Chinese) approach CR through clocks and light signals. The two booklets expand the examples for a general audience.

Introduction to Complex Relativity develops the real slice and the mass shell in detail, with worked calculations for physics students.

Complex Relativity: Full Theory gives the restricted gravitational action, the family of C sectors and the assumptions behind the matter example.

References

[1] Complex Relativity project. *Complex Relativity: Full Theory* and *Introduction to Complex Relativity*, version 1 (2026). github.com/Gauss-K/Complex-Relativity.

[2] J. Ehlers, F. A. E. Pirani and A. Schild, *The geometry of free fall and light propagation*, in *General Relativity: Papers in Honour of J. L. Synge* (1972), pp. 63–84. Reprinted in *Gen. Relativ. Gravit.* **44**, 1587–1609 (2012). doi:10.1007/s10714-012-1353-4.

[3] R. Sachs, *Gravitational waves in general relativity. VI. The outgoing radiation condition*, *Proc. R. Soc. Lond. A* **264**, 309–338 (1961). doi:10.1098/rspa.1961.0202.

[4] C. Cheung, A. Helset and J. Parra-Martinez, *Geometric soft theorems*, JHEP **04**, 011 (2022). arXiv:2111.03045.